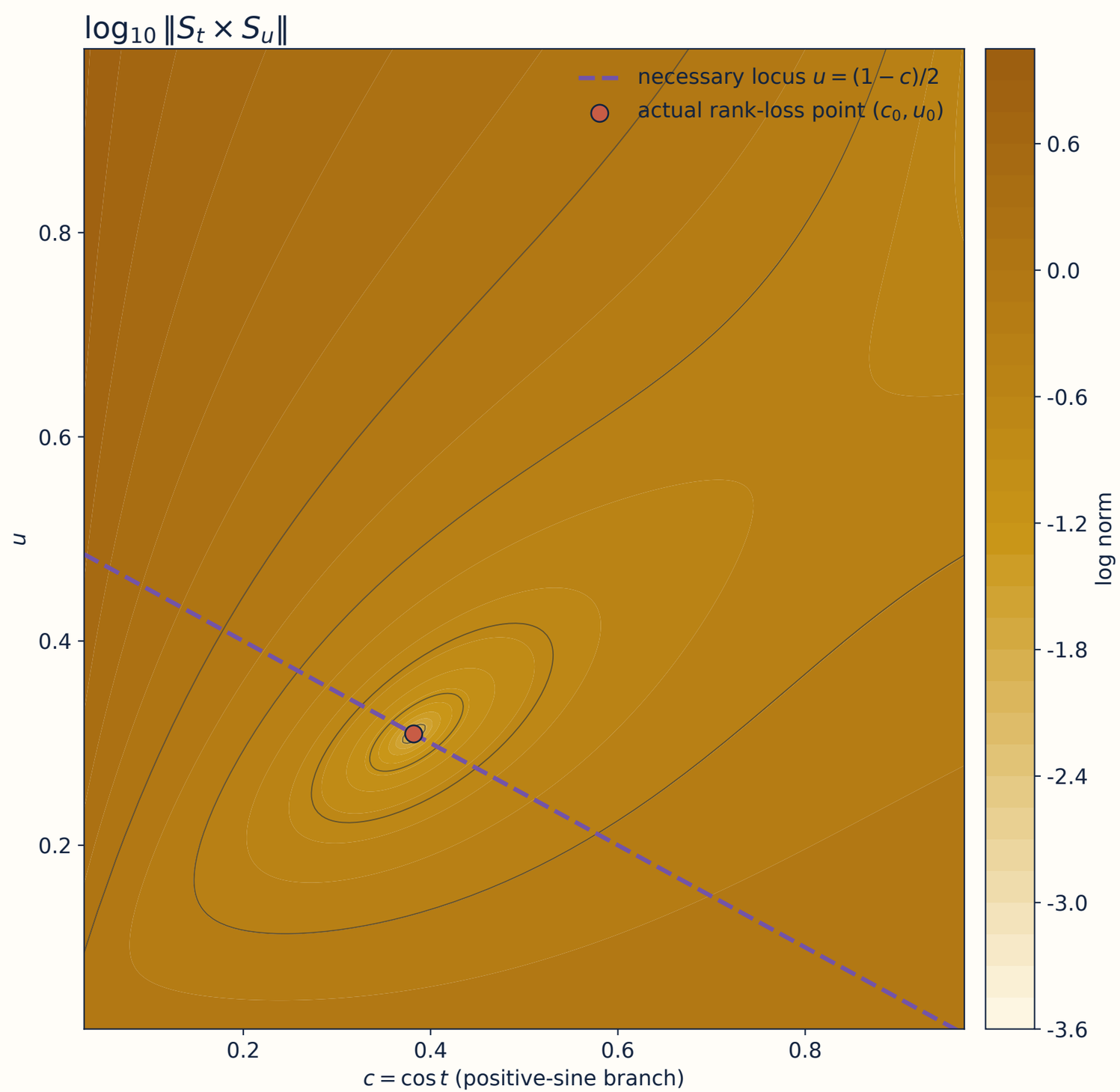
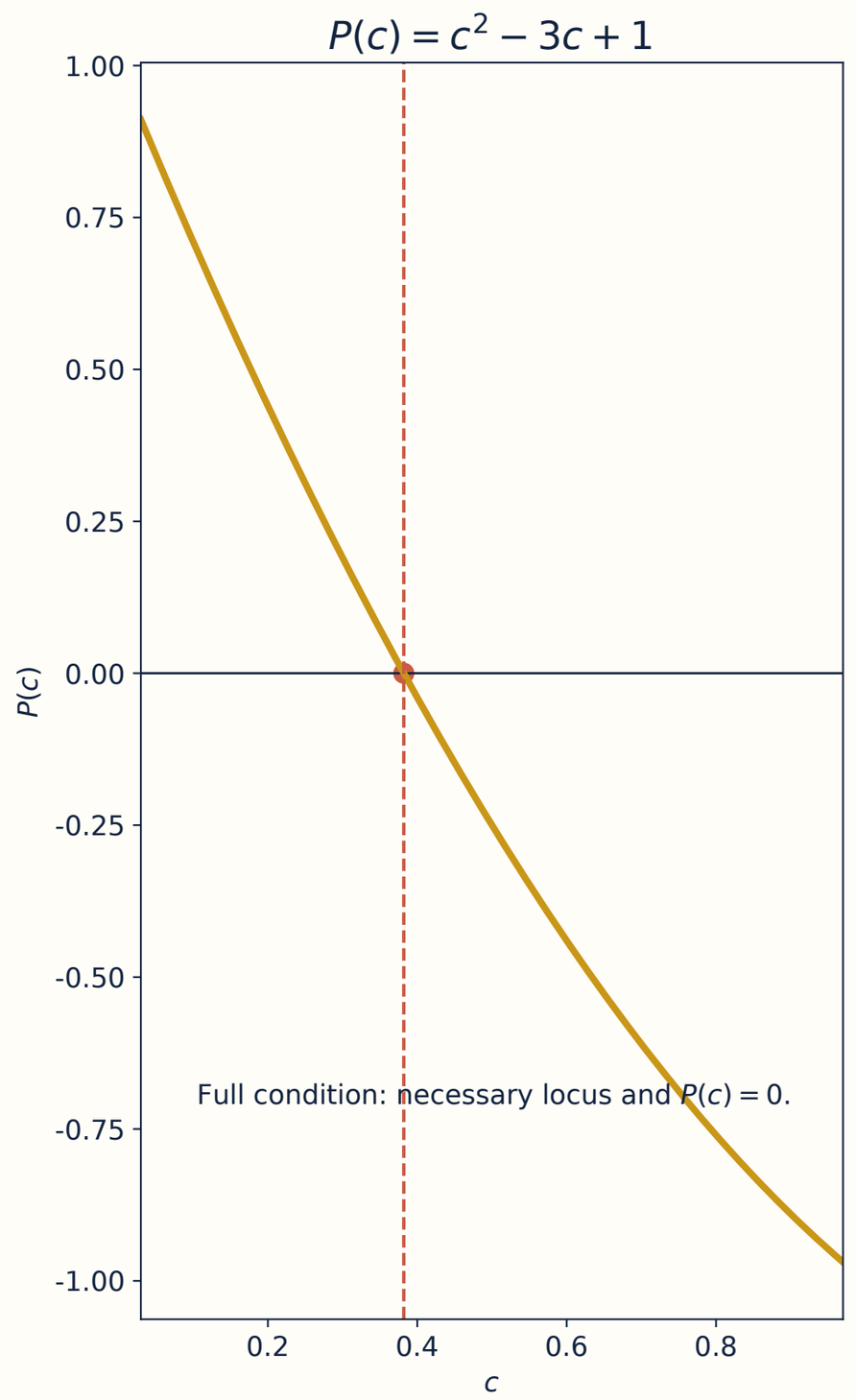

$\log_{10}\|S_t \times S_u\|$
necessary locus $u = (1-c)/2$
actual rank-loss point (c_0, u_0)
u
0.8
0.6
0.4
0.2
0.2
0.4
0.6
0.8
$c = \cos t$ (positive-sine branch)
0.6
0.0
-0.6
-1.2
-1.8
-2.4
-3.0
-3.6
log norm


$P(c) = c^2 - 3c + 1$
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
$P(c)$
Full condition: necessary locus and $P(c) = 0$.
0.2
0.4
0.6
0.8
c